RESEARCH ARTICLE

The Effect of Altering Knee Position and Squat Depth on VMO:VL EMG Ratio During Squat Exercises

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Abstract

Background. Patellofemoral pain syndrome is an extremely common condition, believed to be caused by altered activation of vastus medialis obliquus (VMO), leading to maltracking of the patella. **Aim.** This study aimed to investigate the effect of altering knee movement and squat depth on the ratio of VMO and vastus lateralis (VMO:VL) during squat exercises. **Method.** Eighteen (7 male and 11 female) healthy, asymptomatic participants performed semi-squat exercises with three squat depths (20°, 50° and 80° of knee flexion) while following three knee movement paths (neutral, varus or valgus). Normalized VMO:VL ratio from linear envelope surface electromyography was analysed. **Results.** No significant effect was found for gender ($p = 0.87$), leg dominance ($p = 0.99$) or knee position ($p = 0.44$). A significant effect was found for squat depth ($p < 0.001$) with both the 50° and 80° squats showing increases in VMO:VL ratio ($p = 0.031$ and $p = 0.028$), respectively. The VMO:VL ratio was not influenced by gender, leg dominance or knee position in semi-squat exercises. **Discussion and Conclusion.** Increases in relative VMO activation did occur in 'deeper' squat depths (50° and 80° knee flexion) compared with the 20° condition. Further research is needed in this area concerning the effects of such exercise modifications on a symptomatic patellofemoral pain syndrome population. Copyright © 2015 John Wiley & Sons, Ltd.

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Keywords

patellofemoral joint; squat exercises; vastus medialis obliquus; vastus lateralis

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Introduction

Disorders of the patellofemoral joint (PFJ), commonly classified as 'patellofemoral pain syndrome' (PFPS), are one of the most common knee pathologies to present in both general and sporting populations (Crossley *et al.*, 2000). This condition presents, clinically, as diffuse peri-patellar or retro-patellar pain, aggravated by activities such as prolonged sitting, stair climbing, kneeling, and squatting (Coqueiro *et al.*, 2005). While there is no definite consensus regarding the aetiology and pathophysiology of PFPS (Monteiro-Pedro *et al.*, 1997) the most widely accepted hypothesis is the patellar 'maltracking' theory (Ryan and Rowe, 2006).

The stability and normal movement, 'tracking,' of the PFJ is predominantly maintained by the dynamic balance between the vastus medialis obliquus (VMO) and vastus lateralis (VL) muscles (Ng *et al.*, 2008). The relative activation of these muscles can be measured using

electromyography (EMG) (Cerny, 1995). It has been shown that the VMO/VL amplitude ratio can be used as a highly reliable parameter during single-leg squatting in healthy individuals (Mostamand *et al.*, 2013). It is proposed that, when this muscular equilibrium between VMO and VL is upset, the forces acting upon the PFJ are altered, leading to the 'maltracking' of the patella within the trochlear groove (Rice *et al.*, 1995). The most common imbalances are the result of changes in the contraction and activation of the VMO, resulting in a lateral displacement of the patella as the VMO is unable to counteract the lateral force vector of the VL and iliotibial band (Panagiotopoulos *et al.*, 2006). This lateral maltracking is thought to result in abnormal joint stress and subsequent articular cartilage wear (Powers, 1998).

To counter this, it has been widely proposed that VMO should be preferentially strengthened (i.e. a VMO : VL ratio greater than 1) to increase the medial dynamic stabilization of the patella (Panagiotopoulos *et al.*, 2006). In order to selectively strengthen VMO, previous researchers have posited that altering the biomechanics, such as the position and orientation of the hip (Herrington *et al.*, 2006), knee (Lam and Ng, 2001), ankle (Zakaria *et al.*, 1997) or foot (Gregersen *et al.*, 2006), during quadriceps exercises may preferentially recruit and increase VMO activity. The addition of hip abduction (Coqueiro *et al.*, 2005) or adduction (Boling *et al.*, 2006) has also been suggested to promote preferential VMO recruitment. These results are typically attributed to an anatomical basis, with all muscles of the quadriceps complex innervated by the femoral nerve, which is making preferential activation of VMO quite difficult (Hubbard *et al.*, 1997). Another line of research in this area of research is the timing of VMO activation relative to VL, which is not the focus of the current study.

Overall, even though a number of studies related to other biomechanical aspects of hip, knee and ankle positioning, the main focus of the current study is related to depth of squat at different knee positions, which is a novel concept.

Closed-kinetic chain exercises, such as squats, are commonly recommended for PFPS rehabilitation as such exercises supposedly promote functional stability (Irish *et al.*, 2010) with deceased compression and shear forces acting through the tibiofemoral joint (Witvrouw *et al.*, 1996, Hartmann *et al.*, 2013). Even though it has been shown that training with deep squats elicits favourable adaptations on knee extensor muscle size and function compared with training with shallow squats (Bloomquist *et al.*, 2013), this type of exercises and their effects on preferential activation of VMO muscle are poorly described in the literature. Anecdotal clinical evidence dictates that a 'knees over toes' position is commonly used; however, no evidence exists to confirm this observation. This study was developed to investigate the effects of altering squat depth and knee position on VMO : VL ratio during a quads-based squat in an asymptomatic population. This provides a physiological model of the effects, which could be used as a feasibility study for clinical investigations on patients with PFJ conditions. It was hypothesized that there would be differences in VM/VL ratio between each knee angle and at different squat depth.

Method

Participants

Eighteen subjects (see Table 1 for participant characteristics) met the following inclusion criteria and volunteered to participate in this study.

- Between 18 and 55 years of age
- Able to perform light to moderate exercise for 15 minutes
- Able to understand instructions in spoken English
- No musculoskeletal or neurological disease or disorder affecting normal functioning of the hip, knee or ankle joints

Table 1. Summary of major participant characteristics

	n	Age (years)	Height (cm)	Weight (kg)	Dominant leg (*n*)
Male	7	22.57 ± 1.40	175.86 ± 5.27	71.71 ± 5.38	(R) 4, (L) 3
Female	11	22.27 ± 2.72	165.82 ± 6.08	60.36 ± 8.46	(R) 10, (L) 1
Total	18	22.39 ± 2.25	169.71 ± 7.54	64.78 ± 9.21	(R) 14, (L) 4

(R), right; (L), left.

Note: Values are mean ± standard deviation.

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All participants provided written, informed consent prior to the study. All experimental procedures were conducted according to the Declaration of Helsinki with ethical approval from the Monash University Human Research Ethics Committee (MUHREC).

Electrode placement

Skin over the anterior thigh was lightly abraded with medical grade sandpaper tabs and cleaned with isopropyl alcohol swabs to reduce skin impedance. A bipolar method was used, over the area of greatest muscle bulk of VMO and VL in line with the muscle fibres, on both lower limbs (Figure 1).

The VMO electrodes were placed with the knee in a slightly flexed position, as this increases the confidence that it is actually the VMO being recorded (Earl *et al.*, 2001). Ground electrodes were positioned over the tibial tuberosities, and all electrodes and wires were secured with adhesive hypoallergenic tape.

Instrumentation

The EMG data were collected using pre-gelled 30-mm Ag–AgCl multi-purpose surface electrodes (Skintact, Innsbruck, Austria) with a 30-mm inter-electrode distance. A PowerLab 8/30 system with attached Dual Channel Bio-amplifiers (ADInstruments, Bella Vista, Australia) and LabChart 7.2 software (ADInstruments, Bella Vista, Australia) was used to display and record EMG data. The surface electromyography (sEMG) was band-pass filtered between 10–500 Hz, with a 50-Hz notch, and sampled at 1 KHz.

The squat exercises were performed on a specially designed platform of 7-cm height. This apparatus involved a set of analogue scales (right leg) and a platform step of equal height (left leg) covered with grip tape for safety. The required foot position and arrows designating the required 'path' of knee movement were marked on the platform in electrical tape.

Procedures

After the electrodes were attached, the subjects were seated in a podiatry chair, barefoot, in a supported position in approximately 90° of hip flexion and 45° of knee flexion. The stability of subjects was achieved using straps to fix thigh and trunk. Subjects were instructed to keep their hands over their lap while they were performing three alternate knee extensions. Subjects then performed a maximal voluntary isometric contraction (MVIC) at 45° of knee flexion for each leg, against resistance provided by the researcher, using hands on their shin and receiving constant verbal encouragement throughout. MVIC data were collected from the left leg first in all subjects.

Subjects then stood to have their weight measured before being positioned on the platform apparatus as per the markings and instructions provided (Figure 2).

They were asked to ensure that half of their body weight (±2.5 kg) was maintained over their right leg (over the scales) for the remainder of the tasks. The

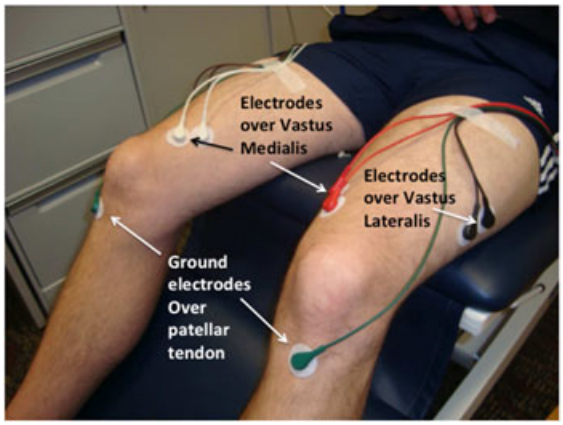


Figure 1. Example showing electrode placement on a sample subject with wires attached and taped into position. Note: electrodes have not been attached with tape to aid in clarity of positioning

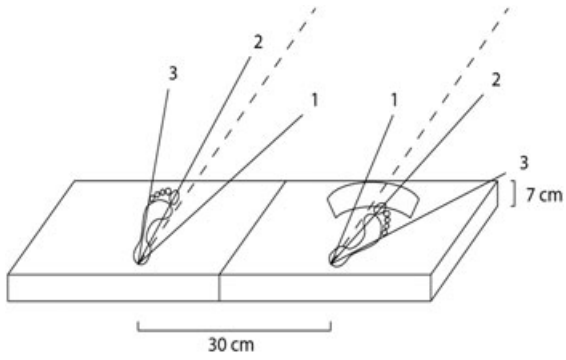


Figure 2. Diagram of foot position during semi-squat tasks showing feet positioned on the lines provided in a 7° externally rotated position. Apparatus had (R) foot on a set of analogue scales and (L) foot on a step of matching height. Lines were marked for each knee 'position'. Line 2 was 'neutral' (knees over big toe in 7° external rotation), Line 1 was 20° valgus and Line 3 was 20° varus. Feet were positioned 30 cm apart at the heel

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subjects were then asked to maintain a relaxed, static standing posture so that a baseline standing sEMG value could be obtained.

Subjects performed squats following each of the three knee positions (varus, valgus and neutral paths, i.e. 'knees out', 'knees in' or 'knees over toes', respectively) for each squat depth (20°, 50° and 80° of knee flexion) for a total of nine squats with 60-second intervals. The order in which the squats were performed was assigned by means of an atmospheric noise-based random number generator (random.org, Dublin, Ireland) to limit the effects of training and fatigue. The performance of each squat was controlled with standardized instructions based on a generic stopwatch. Each squat consisted of a 3-second descent ('Down, Down, Down'), a static contraction for 5 seconds ('Hold, Hold, Hold, Hold, Hold') and finally a 3-second ascent ('Up, Up, Up'). All squats were performed without shoes and with the arms in a comfortable position across the chest. To control the squat movements in sagittal plane, the participants were instructed to perform the movements in neutral position, and avoid movements at more forward (bringing the knee forward over the toes) or backward (sit back more as the knee is flexed) positions. Similarly, foot pronation and supination movements were controlled by monitoring of the barefeet over the platform. Any squat movement that resulted in the clearance of one side of the foot from the platform was disqualified, and the participants were instructed to repeat that movement.

Measurements of knee flexion, and thus squat depth, were taken with an international standard 32-cm universal goniometer. The goniometer was aligned to specific anatomical landmarks (Norkin and White, 1995), with the fulcrum of the goniometer aligned over the lateral epicondyle of the femur, the proximal arm aligned with the greater trochanter of the femur and the distal arm with the lateral malleolus of the fibula. The assessor kept the goniometer aligned with the aforementioned bony landmarks during the active squat.

The squat depths chosen were based upon knee positions commonly found in everyday functional activities in which the quadriceps act to produce movement of stability. The decision to test several squat depths was based on the theory that the length of the muscle will influence the tension (force) it is able to produce (Rassier *et al.*, 1999) and thus may result in changes in the relative activation of VMO and VL. The first depth of 20° of knee flexion was chosen based on the 'normal' knee joint positions during gait. This position is between the 30° of knee flexion present during toe-off and the 5° of flexion during mid-stance (Levangie and Norkin, 2006). The 80° of knee flexion was chosen for the 'deep' squat depth as this position is demonstrated at the 'weight acceptance' phase of stair ascent (Levangie and Norkin, 2006) before pull-up. The 80° of flexion was chosen for the deepest squat as lower limb muscle activity peaks at 70–90° of knee flexion (Sczepanski *et al.*, 1991), and squat depths greater than this show significant stress on the structures of the knee joint (Escamilla *et al.*, 2001). The final squat depth, of 50° knee flexion, was chosen as it offers a good 'spread' of testing positions and falls outside the reported 10° of measurement error associated with goniometry (Edwards *et al.*, 2004).

All semi-squats were performed with the feet positioned in 7° of external rotation as this is the normal foot angle during gait (Murray, 1967) and is within the range of 5–18° of lower limb external rotation, which is optimal for balance in standing and during gait (Magee, 1992).

Once all squats had been completed, another standing resting baseline was recorded. The participants' dominant leg was determined by asking which leg he or she would use to kick a football (Zeller *et al.*, 2003).

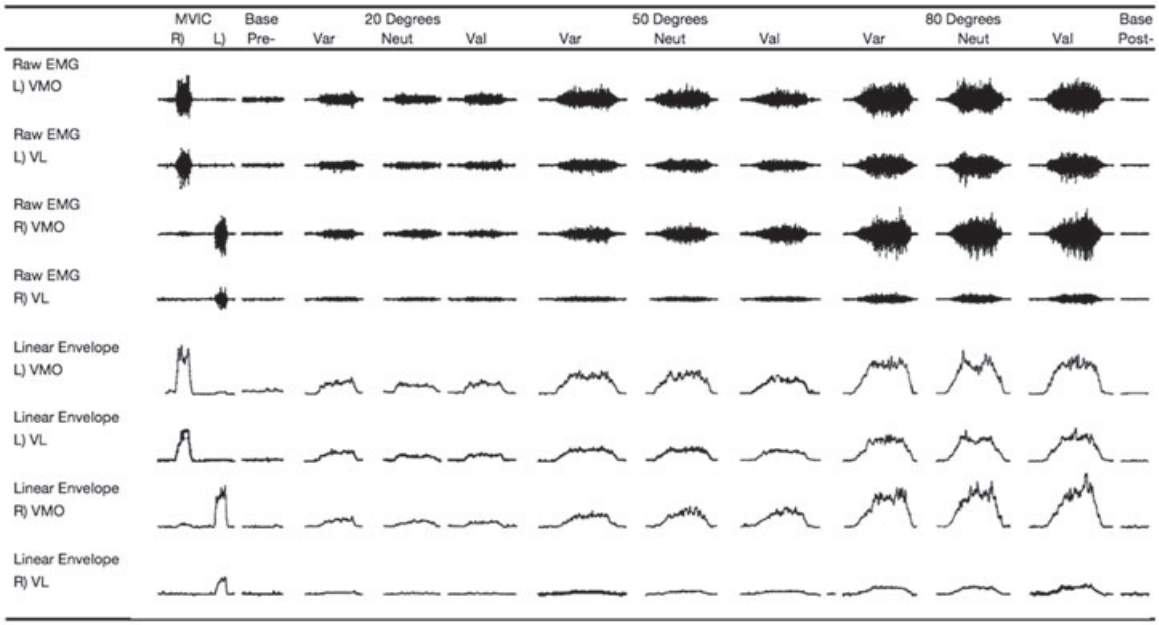
Data analysis

The raw sEMG data were transformed via full wave rectification (absolute) followed by low-pass filtering before being averaged over time during the 5-second static contraction during the squats. EMG normalization was then performed based on the recorded average values for the MVIC for each muscle of each limb (Figure 3).

The ensemble-averaged sEMG data were then analysed using the Statistical Package for Social Sciences (SPSS) version 19.0.0 (IBM, New York, USA). Differences in mean VMO : VL ratio were analysed using a multifactorial analysis of variance (ANOVA) with four factors: sex, side (dominant or non-dominant), squat depth (20°, 50° or 80° of knee flexion) and knee position (varus, valgus or neutral). The critical alpha level was set at $\alpha = 0.05$.

The activity of the VMO and VL was presented as a ratio of relative activation as these two muscles are antagonistic to each other on patellar tracking, and

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Abbreviations: VMO = vastus medialis obliquus; VL = vastus lateralis; MVIC = maximum voluntary isometric contraction; Var = varus; Val = valgus; Neut = neutral; Pre- = pre-test; post- = post-test

Figure 3. Example electromyography recordings, both raw and linear envelopes, from one participant

thus their effects may be neutralized (Cerny, 1995). All EMG data were normalized to accommodate individual subject characteristics such as muscle fibre type and size and variations in muscle shape (Burden, 2010), and thus allowing for comparisons to be made within and between subjects and studies (Lehman and McGill, 1999).

Differences in relative muscle activation between the sexes were analysed in this study, as significant differences are known to exist between the male and female body.

The effect of leg dominance on VMO : VL ratio was also analysed in this study. Leg dominance was assessed to determine if these potential 'imbalances in muscular strength and recruitment patterns' would affect the VMO : VL ratio of the participant.

Results

There was no statistically significant difference in VMO : VL ratios found between male and female participants (F (1.6) = 0.028, $p = 0.872$) for leg dominance, squat depth and knee position. Similarly, no statistically significant difference existed in VMO : VL ratios between dominant and non-dominant lower limbs of the participant population (F (1.17) < 0.001, $p = 0.994$). Based on this, the data from both sexes and both dominant and non-dominant limbs were pooled for all further analysis.

The data for overall activity level, expressed as a percentage of the MVIC, and VMO : VL ratios for each condition are presented in Table 2.

The differences between the muscle activations for shallow squats compared with the deepest ones are higher for valgus (VMO: 57.33 − 10.07 = 47.26, VL: 72.27 − 18.20 = 54.07) compared with movements in varus (VMO: 46.53 − 10.55 = 35.98, VL: 64.43 − 17.86 = 46.57) positions for both VMO and VL muscles.

Analysis by multifactorial ANOVA (squat depth and knee position) revealed that the interaction of squat depth and knee position was significant (F (2.7, 45.9) = 5.97, $p = 0.002$, $\eta^2 = 0.26$). *Post hoc* pairwise analysis using independent t-tests (Table 3), showed no significant difference between any two conditions ($p > 0.02$, $\alpha = 0.006$).

The ANOVA demonstrated no statistically significant effect for knee position (F (2.34) = 0.84, $p = 0.44$, $\eta^2 = 0.05$) but did show a significant effect for squat depth (F (2.34) = 28.40, $p < 0.001$, $\eta^2 = 0.63$) with both the 50° and 80° squats showing increases in VMO : VL ratio ($p = 0.031$ and $p = 0.028$, respectively) compared with the 20° squat with pairwise comparison. Analysis of pre-test and post-test standing baseline measures of VMO : VL ratio showed no significant difference ($p = 0.49$).

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Table 2. Summary of muscle activation for each condition for vastus medialis obliquus (VMO) and vastus lateralis (VL), expressed as a normalized percentage of maximum voluntary isometric contraction and VMO : VL ratio

Squat depth (°)	Knee position	VMO[a]				VL[a]				VMO : VL ratio			
		μ	SD	Min	Max	μ	SD	Min	Max	μ	SD	Min	Max
20	Varus	10.55	7.40	3.28	30.34	17.86	19.50	4.78	119.69	0.76	0.34	0.40	1.33
20	Neutral	9.95	7.51	1.12	31.91	17.29	22.36	5.51	138.76	0.73	0.35	0.33	1.35
20	Valgus	10.07	7.45	0.90	28.55	18.20	21.74	7.44	133.80	0.69	0.29	0.38	1.24
50	Varus	25.20	14.07	8.31	69.30	35.42	30.31	15.43	180.92	0.89	0.38	0.48	1.69
50	Neutral	28.95	17.92	6.74	74.45	40.38	44.14	10.33	267.52	0.93	0.39	0.47	1.69
50	Valgus	29.41	19.16	8.52	92.83	40.34	42.76	9.28	254.56	0.93	0.34	0.53	1.56
80	Varus	46.53	23.69	15.03	95.54	64.43	51.77	28.26	315.33	0.86	0.35	0.42	1.54
80	Neutral	56.23	32.29	16.76	146.16	76.72	82.47	21.54	512.68	0.91	0.38	0.48	1.84
80	Valgus	57.33	32.30	20.18	139.98	72.27	58.52	20.38	325.08	0.96	0.38	0.54	1.82

VMO, vastus medialis obliquus; VL, vastus lateralis; μ, mean; SD, standard deviation; Min, lowest ensemble-averaged, normalized result; Max, highest ensemble-averaged, normalized result; MVIC, maximum voluntary isometric contraction.

Note: all values expressed are from pooled group data irrespective of sex or leg dominance.

[a]Values for VMO and VL are expressed as percentage of MVIC (normalized values).

Table 3. Analysis of variance summary table for effects of sex, leg dominance, squat depth and knee position and interactions among the variables

Condition	*n*	df	SS	MS	Mauchly's (Sig)	*F*	Sig	η^2
Sex[b]	18	1	0.031	0.031	—	0.028	0.872	—
Error	18	6	6.585	1.097	—	—	—	—
Leg dominance[b]	18	1	6.049^−5	6.049^−5	—	<0.001	0.994	—
Error	18	17	19.594	1.153	—	—	—	—
Squat depth	18	2	1.272	0.636	0.866	28.4	<0.001	0.626
Error	18	34	0.762	0.022	—	—	—	—
Knee position	18	2	0.020	0.010	0.053	0.836	0.442	0.047
Error	18	34	0.402	0.012	—	—	—	—
Squat depth × knee position	18	4	0.133	0.033	0.024	5.969[a]	0.002[a]	0.26[a]
Error	18	45.91[a]	0.380[a]	0.008[a]	—	—	—	—

n, number of participants; df, degrees of freedom; SS, type III sum of squares; MS, mean square; Mauchly's (Sig), significance values for Mauchly's test of sphericity; *F*, *F*-statistic; Sig, significance of *F*-statistic; η^2 = partial eta squared; —, not applicable.

[a]Values corrected using Greenhouse–Geisser estimation as Mauchly's test is significant.

[b]As the effect of both sex and leg dominance were both non-significant, data for these groups were pooled.

Discussion

The results of this study showed no significant difference in VMO : VL ratio when comparing semi-squat exercises in neutral ('knees over toes'), varus (20° outwards movement path) and valgus knee positions (20° inwards path). As far as the authors are aware, this is the first time that the effect of altering knee movement and position on VMO : VL ratio during semi-squat exercises has been investigated. Anecdotal evidence indicates that, for such exercises, participants are most commonly taught squats with 'knees over toes', with the foot in a slightly externally rotated position. This neutral position may be used because the 'correct' biomechanical position is thought to result in improved patellofemoral kinematics; however, this is simply speculation and should be treated as such until new evidence is available.

The results of this study also demonstrated that a significant effect for squat depth existed, with both the 50° and 80° squat depths showing significant increases in VMO : VL ratio compared with the 20° squat depth. Such findings are in agreement with the previous work of Anderson and colleagues (Anderson *et al.*, 1998) who found that significant differences existed between VMO : VL ratios obtained during semi-squat exercises to a depth of 30° and 90° knee flexion ($p < 0.01$) in a 'narrow' stance position (feet and hip width apart). A significant difference also existed between both 30° and 60° ($p < 0.02$)

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and 30° and 90° ($p < 0.01$) squat depths in a wide stance position (feet twice hip width apart). Similar results have been obtained by the work of Isear who reported significant differences in muscle activity existed when comparing 0–30°, 30–60° and 60–90° movement arcs during a squat exercise; however, no *p*-value was reported (Isear *et al.*, 1997). Mirzabeigi and colleagues found no preferential VMO activation ($p > 0.05$) or changes in muscle activity between 30–16° and 15–0° arcs during semi-squats (Mirzabeigi *et al.*, 1999); however, one should take into account the 'shallow' squat depth used when considering these results. Several studies have also investigated different degrees of knee flexion for isometric knee extension (Cerny, 1995, Lam and Ng, 2001); however, none of these studies found any evidence of changes in VMO : VL ratio or in VMO and VL activation.

No statistically significant difference was demonstrated between male and female participants in the present study, with such results in agreement to several similar studies on the preferential activation of VMO (Willis *et al.*, 2005, Herrington *et al.*, 2006). Similarly, no significant difference was found when comparing the values obtained from the dominant and non-dominant legs of the participants. This is, to our knowledge, the only reported analysis of the effect of leg dominance on VMO : VL ratio.

The VMO : VL ratios obtained from this study ranged from 0.69 to 0.96 during exercise conditions. These values are similar to previous reported results 0.86 (Mostamand *et al.*, 2013) during single-leg squatting down to 45° knee flexion. However, they are lower than the 'normal' 1:1 ratio reported by Souza and Gross (Souza and Gross, 1991), and those reported by Cerny (Cerny, 1995) of 1.0–1.2 from healthy subjects performing closed kinetic chain exercises, yet are higher than the value of 0.54 reported by Powers (Powers, 2000) for healthy subjects. It should be noted that the VMO : VL ratios obtained in the current study were not reached levels reported by Souza and Gross (1991) and Cerny (1995). Weaknesses in control of methodological issues such as standardizing the position of the feet, accounting to limb dominance and gender, and providing feedback on weight-bearing asymmetry, could be considered as reasons for these discrepancies.

Limitations of study

Our present study has some limitations: Firstly, the data were obtained from a healthy population with no pain history; therefore, the results may not necessarily be extrapolated to patients with PFPS. Secondly, the effects were evaluated in younger participants (less than 55 years). Older individuals may respond differently to the experimental conditions. Thirdly, the results in this study should be evaluated based on the existence of non-linear relationship between EMG and muscle force (Hof 1997). It should be also noted that, because of the multiple comparisons that were made in this study, calculation of sample size was not feasible. This should be considered as a limitation for this study. Another limitation is that, even though EMG was normalized, the inter-subject variability in surface EMG is always an issue, particularly during dynamic tasks due to skin movement over underlying motor units. Finally, there are a number of statistical comparisons relative to the sample size, which increases risk of type 1 error, which should be also considered as a limitation.

Clinical relevance of the findings

In general, the results of this study, and of previous studies, demonstrate that VMO has increased levels of activation, relative to VL, when semi-squat exercises are performed to a 'mid-range' (i.e. 40–90°) of knee flexion, as opposed to an early range (0–30°) depth. It is essential to note, however, that none of the conditions tested resulted in preferential VMO activation (i.e. a VMO : VL ratio greater than 1). Theoretically, training with exercises that result in a VMO : VL ratio lower than 1:1 will result in greater strength gains in the VL muscle, potentially increasing the lateral force acting on the patella and thus symptoms.

What this study did not take into account was the actual *magnitude* of muscle activation. While there was no significant difference found between a squat to 50° compared with 80° in terms of VMO : VL ratio, one would expect that the actual magnitude of contraction to be significantly higher with increased depth and thus increased loading. This would then seem to suggest that a deeper squat, while having no benefits in terms of preferential VMO activation, would result in overall increased training effects through the overload principle, when compared with a 'shallower' squat.

The magnitude of contraction and the VMO : VL ratio are not, however, the only two factors that must be

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considered before utilizing the results of this study. Another important consideration is the joint reaction force and stress produced by an exercise. As squat depth increases, so does the compressive load on the PFJ (Escamilla *et al.*, 2001), thus symptomatic individuals may find it difficult to perform a deeper squat because of pain provocation. It has been reported that the best range of motion in which to train quadriceps strength, while incurring minimal joint reaction force and stress is between 0–50° of knee flexion during closed kinetic chain exercises (Grelsamer and Klein, 1998, Escamilla *et al.*, 2001).

While this study is focused towards the rehabilitation of PFPS, it is impossible to accurately and safely generalize results from a healthy sample to a symptomatic population. It has been shown that individuals with PFPS commonly experience pain during functional activities such as ascending and descending stairs, squatting and prolonged sitting (Crossley *et al.*, 2000). What these activities have in common is a functional range of knee flexion between 0–40° (Liaw, 2000). What is also recommended is that the main goal of PFPS rehabilitation is to maximize quadriceps strength while minimizing PFJ reaction force and stress (Tang *et al.*, 2001) and that such training should occur in a closed kinetic chain position as it promotes functional stability (Miller *et al.*, 1997, Irish *et al.*, 2010).

Conclusions

This study found that the relative activation of VMO and VL was not influenced by gender, leg dominance or knee position in semi-squat exercises. Increases in relative VMO activation did occur in 'deeper' squat depths (50° and 80° knee flexion) compared with the 20° condition; however, no one condition resulted in preferential VMO activation (i.e. VMO : VL ratio >1).

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